

**Example 8:** Raymond Smullyan posed many puzzles about an island that has two kinds of inhabitants, Knights, who always tell the truth, and their opposites, Knaves, who always lie. You encounter two people A and B. What are A and B if A says "B is a Knight" and B says "The two of us are opposite types"?

**Soln:** Suppose, A is a Knight. Since A always speaks the truth, so B is a Knight. Since B always speaks the truth, it is true that both A and B are of opposite types. Since, Both A and B are Knights, this is a contradiction.

~~Suppose~~ **∴ A is a Knave**. Since A always lies, B is not a Knight and hence is a Knave. The fact that B is a Knave is consistent with the statement "The two of us are opposite types" being False, since both A and B are Knaves.

### Muddy children puzzle

**Example 9:** A father tells his two children, and a boy and a girl, to play in their backyard without getting dirty. However, while playing, both children get mud on their foreheads. When the children stop playing, the father says "At least one of you has a muddy forehead." and then asks the children to answer "Yes" or "No" to the question: "Do you know whether you have a muddy forehead?" The father asks this question twice. What will the children answer each time this question is asked, assuming that a child can see whether his or her sibling has a muddy forehead, but cannot see his or her own forehead? Assume that both children are honest and that the children answer each question simultaneously?

**Soln:** On the first time, both the boy and the girl will answer No. This is because, both will see each other's forehead and see mud on it. This proves the statement "At least one of you has a muddy forehead" correct, however they cannot conclude <sup>whether</sup> they have mud on their own forehead or not.